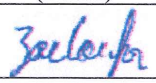
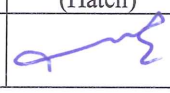
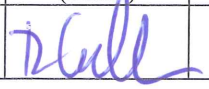
MA'ADEN معادن

MA'ADEN ENGINEERING MANUAL

Ground Improvement By Vibro-Replacement

MD-101-SMEM-EG-CV-SPC-0011

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Table of Contents

1. Introduction 3

1.1 Scope 3
1.2 Purpose 3

2. Key Terms and Definitions 3

3. Referenced Documents 4

3.1 Statutory Regulations 4
3.2 Industry Codes and Standards 4
3.2.1 Local Codes and Standards 5
3.2.2 International Codes and Standards 5
3.2.3 Alternative Codes 6
3.3 Ma'aden Standards 6

4. Submittals 6

5. Performance Criteria 7

6. Materials and Equipment 7

6.1 Stone Columns 7
6.2 Vibrator 8

7. Execution 8

7.1 Protection of Facilities and Utilities 8
7.2 Site Inspection and Layout of Work 8
7.3 Runoff Management 8
7.4 Field Quality Control 8
7.5 Field Testing 9
7.6 Restrictions 10

8. Quality Control 10

1. INTRODUCTION

1.1 Scope

This document specifies the minimum requirements for ground improvement using the Vibro Replacement (VR) method, for Ma'aden projects. The intent of the VR specified herein is to provide soil improvement and reinforcement within the limits indicated on project drawings to achieve the performance criteria of these specifications.

VR is a stone column installation method in which a Vibrator (down-hole vibratory probe) compacts aggregate into a column and densifies the surrounding soils to increase bearing capacity and reduce settlement of a planned structure. VR can also serve to increase shear resistance, particularly when a load is placed above the columns.

The method is generally applicable to soils, with greatest benefits observed in fine-grained, mixed and layered soils which do not densify easily when subjected to vibration alone. The VR technique was developed specifically for these soils, effectively extending the range of soil types that can be improved with the deep vibratory process. For coarser grained soils such as sands, alternative and simpler vibration methods might be considered.

Situations in which the VR technique may not be applicable include soil layered with peat, loose collapsible fill with voids, and obstructions to penetration. Additionally, the design engineer should take account of the enhanced permeability provided by the stone columns, and consequent surface water penetration to underlying soil layers.

This document is intended to be used in conjunction with the referenced documents listed herein.

1.2 Purpose

The purpose of this specification is to set the minimum engineering standard for ground improvement using VR methods.

This is a generic document. A Project specification must be prepared by the designing engineer in compliance with this Master Specification.

2. KEY TERMS AND DEFINITIONS

Contractor	An entity, engaged or approved by the Owner, entrusted with a contract to perform work or cause work to be performed.
CPT	Cone Penetration Test
Dry, Bottom Feed Method (Displacement)	This method employs the same type vibrator as Wet, Top Feed Method, but with the addition of a hopper and supply tube to feed the stone backfill directly to the tip of the vibrator. Bottom feed Vibro Replacement is a generally dry operation with little spoil, extending the range of sites that can be treated. Treatment to depths greater than 24 m is possible and is not inhibited by the presence of groundwater. The stone backfill is densified in 0.6 to 0.9 m lifts. Typical setups include VR unit suspended from a crane, mounted on an excavator, or leader mast mounted.

Employer	Saudi Arabian Mining Company (Ma'aden) and its authorized representatives and respective affiliates.
FQCR	FQCR or Field Quality Control Representative is the Employer's representative given specific inspection tasks identified in this specification. The FQCR shall work for and be paid by the Employer.
SPT	Standard Penetration Test.
Stone Columns	Columns of aggregate (crushed rock or gravel of suitable grade) used to create a composite soil structure with improved load capacity.
Wet, Top Feed Method (Replacement + Displacement)	With this method, jetted water is used to remove soft material, stabilize the vibrator hole, and ensure that the stone backfill reaches the tip of the vibrator. This is the most commonly used and most cost-effective of the deep vibratory methods. Wet spoil is generated with this method, and must be carefully managed when working on confined sites or in environmentally sensitive areas. The stone backfill is densified in 0.6 to 0.9 m lifts.
Work	The entire work and Services and all things which the Contractor is or may be required to execute, supply and do under the Contract having regard to the Scope of Project, or to cause to be executed, supplied or done, for the purposes of constructing and Commissioning the Plant.

3. REFERENCED DOCUMENTS

This specification requires compliance with the provisions of the latest revision of the relevant standards or codes of the following authorities and technical organizations, as applicable, and specifications which shall be considered to form part of this specification, where applicable.

3.1 Statutory Regulations

All Work performed and materials furnished shall comply with the applicable statutory regulations, codes and other requirements of the Kingdom of Saudi Arabia (including but not limited to directives of the High Commission for Industrial Security, Royal Commission regulations applicable to the Site, regulations of the Presidency of Meteorology and Environment, Saudi Building Code and other applicable codes and regulations). The Contractor shall ensure that he is fully aware and informed of all Government laws, local ordinances, regulations and all orders and decrees of bodies or tribunals having any jurisdiction or authority over the Project and/or Site, which in any manner affect the Project and those engaged or employed on the Project, or which in any way affect the conduct of the Work. The Contractor shall at all times observe and comply with all such Government and/or Ministry laws, bylaws, ordinances, codes, regulations, orders and decrees.

3.2 Industry Codes and Standards

All equipment and systems shall conform to the following.

3.2.1 Local Codes and Standards

SBC303	Soil and Foundations Requirements

3.2.2 International Codes and Standards

3.2.2.1 American Concrete Institute (ACI)

ACI 229	Controlled Low Strength Material (CLSM).

3.2.2.2 American Society for Testing and Materials (ASTM)

ASTM C33	Standard Specification for Concrete Aggregates
ASTM C136	Standard Test Method for Sieve Analysis of Fine and Coarse Aggregates
ASTM D422	Standard Test Method for Particle-Size Analysis of Soils
ASTM D698	Standard Test Method for Laboratory Compaction Characteristics of Soil Using Standard Effort 12,400 ft-lbf/ft3 (600 kN-m/m^3)
ASTM D1140	Standard Test Method for Amount of Material in Soils Finer than the No. 200 (75 µm) Sieve
ASTM D1143	Standard Test Method for Deep Foundations Under Static Axial Compressive Load
ASTM D1194	Standard Test Method for Bearing Capacity of Soil for Static Load and Spread Footings
ASTM D1241	Standard Specification for Materials for Soil-Aggregate Sub-Base, Base and Surface Courses
ASTM D1556	Standard Test Method for Density and Unit Weight of Soil in Place by the Sand-Cone Method
ASTM D1557	Test Method for Laboratory Compaction Characteristics of Soil Using Modified Effort 56,000 ft-lbf/ft3 (2700 kN-m/m^3)
ASTM D1586	Standard Test Method for Standard Penetration Test (SPT) and Split-Barrel Sampling of Soils
ASTM D2487	Standard Practice for Classification of Soils for Engineering Purposes (Unified Soil Classification System)
ASTM D2922	Standard Test Methods for Density of Soils and Soil-Aggregate in Place by Nuclear Methods (Shallow Depth)
ASTM D2937	Standard Test Method for Density of Soil in Place by the Drive-Cylinder Method
ASTM D2940	Standard Specification for Graded Aggregate Material for Bases or Sub-Bases for Highways or Airport

ASTM D3282	Standard Classification of Soils and Soil-Aggregate Mixtures for Highway Construction Purposes
ASTM D 3441	Standard Test Method for Mechanical Cone Penetration Tests of Soil
ASTM D 3740	Standard Practice for Minimum Requirements for Agencies Engaged in the Testing and/or Inspection of Soil and Rock Used in Engineering Design and Construction
ASTM D4253	Standard Test Methods for Maximum Index Density and Unit Weight of Soils Using a Vibratory Table
ASTM D4254	Standard Test Method for Minimum Index Density and Unit Weight of Soils and Calculation of Relative Density
ASTM D4318	Standard Test Method for Liquid Limit, Plastic Limit and Plasticity Index of Soils
ASTM D4428/D4428M	Standard Test Methods for Crosshole Seismic Testing
ASTM D 5778	Standard Test Method for Electronic Friction Cone and Piezocone Penetration Testing of Soils

3.2.3 Alternative Codes

Alternative codes may be used for ground improvement if approved by the Employer.

3.3 Ma'aden Standards

MD-101-SMEM-EG-CV-SPC-0001	Earthworks
MD-101-SMEM-EG-CV-SPC-0008	Topographical survey
MD-101-SMEM-EG-CV-SPC-0009	Geotechnical investigation
MD-101-SGGM-PM-GE-REF-0001	Glossary of Terms, Acronyms and Documents

4. SUBMITTALS

A. The following shall be submitted to the Employer for review:

1. Safety documents.
2. Baseline examination report, including a survey of installations, buildings or structures within 5 m of the VR work areas, and an assessment of construction effects on these structures.
3. Construction drawings showing treatment point locations, column ID numbers, depths and proposed sequence, and adjacencies of existing structures.
4. Equipment list and methodology.

5. The source, gradation and description of proposed stone column aggregate.
6. Field testing and lab testing program.
7. Method statement for confirming installation procedures are meeting the specification.

B. The following shall be submitted to the Employer by the VR Contractor during the work execution:

1. Accurate daily records that include VR number, start/finish time of the VR, depth of treatment, ammeter readings and backfill quantities.
2. Any change in the predetermined VR program necessitated by a change in the subsurface conditions.

C. The following shall be submitted to the Employer by the VR Contractor after the completion of the work:

1. A report documenting the observations and results of the tests. This report will certify that the stone columns have been constructed as per the drawings and the bearing pressure has been achieved within settlement tolerances.
2. As-built drawing showing depicting any significant changes to the shop drawing.

5. PERFORMANCE CRITERIA

The design engineer shall detail the geotechnical design parameters to be achieved when specifying the ground improvement:

- An allowable bearing capacity with a maximum total settlement and differential settlement over a certain length, and any other specific requirement of the project
- The depth of the vibro replacement
- The minimum replacement ratio (area of a VR/footing area or for area treatment, area of a VR/the tributary foundation area for a VR and the minimum VR depth below footing bearing level.

6. MATERIALS AND EQUIPMENT

6.1 Stone Columns

Unless otherwise stated, aggregate shall be used for vibro replacement. The aggregate shall consist of relatively hard, angular to sub-angular durable rock fragments and meet ASTM C33 course aggregate Class S1, with the size of particles in the range of 3 mm to 40 mm. The material to be used should be approved by the Employer. The VR Contractor shall produce evidence that the material to be used meets or exceeds the design requirements. Submittals from the VR Contractor shall be in accordance with Section 4.

6.2 Vibrator

A Vibrator is a down hole vibratory probe producing horizontal vibrations from an energy source located near the tip of the probe. The vibrator shall be a minimum of 400 mm in diameter. Complete equipment specifications shall be submitted to the Employer prior to commencement of the fieldwork.

7. EXECUTION

7.1 Protection of Facilities and Utilities

A. If an adjacent installation, building or structure is within 5 m of the VR work area, a relevant building examination shall be performed prior to initiating the work. The installation, building or structure must also be monitored for settlement, movement or cracking during any work within 5 m of the structure.

B. Monitoring devices shall be capable of reading to an accuracy of 1.5 mm.

C. Readings shall be taken prior, during and after installing the stone columns.

D. Contractor shall stop work and notify the Employer if the installation, building or structure outlined in A above shows any signs of movement.

7.2 Site Inspection and Layout of Work

The Contractor shall accurately lay out center of foundations and stone columns in accordance with the approved drawings.

7.3 Runoff Management

If the Wet, Top Feed method of construction is to be used, sludge and runoff shall be managed within the construction zone, and in accordance with the project environmental and construction requirements.

7.4 Field Quality Control

Inspection and test requirements include the following:

- All VR operations shall be performed under the inspection of the FQCR
- Monitoring and logging of VR operations shall be done by the FQCR
- The FQCR will provide site inspection to ensure performance of the VR work. This inspection may include the following: observance of the VR Contractor's procedures, recording of stone column aggregate quantities, and recording of compaction time.
- A test report in accordance with Section 6 of the stone column aggregate material shall be submitted to the Engineer to establish the suitability
- The FQCR will perform elevation monitoring of existing structures or utilities during performance of any VR within 5.0 m of such structure or utility. If any settlement is observed, the Contractor will be directed to discontinue work within 5.0 m of the affected structure/utility until the engineer determines an alternate procedure to avoid additional settlement.

7.5 Field Testing

A. Penetration Testing, SPT or CPT (if appropriate for soil conditions and specified).

All testing to determine specification compliance shall be provided by the Contractor, paid by the Employer, and shall consist of Penetration Testing and on-site observation during the Vibro Replacement.

1. SPT/CPT testing shall be performed at points which are equidistant and farthest from two, three, or four probes, whichever is applicable, with a multiple probe pattern under a column or wall footing.

2. The work should consider an averaging of SPT/CPT values for the test hole, and also the location of the test hole within the treated area. The required criteria for the production of this work should relate to parameters used in the stone column design as determined by the SPT/CPT work (e.g. to limit settlement and to provide a bearing capacity as shown on the drawings/project specifications and requirements).

3. Areas of the Site not adequately densified as determined by the specified SPT/CPT values shall be re-compacted by Vibro Replacement at no additional cost to the Employer.

4. The Employer will provide Site inspection to ensure proper performance of the Vibro Replacement work. This inspection may include observance of the VR Contractor's procedures, recording of aggregate quantities, and recording of ammeter information.

B. Static Load Testing (if specified).

Testing to determine specification compliance shall be provided by the Contractor, and shall consist of a full scale static load test.

1. The load test shall be performed at an actual foundation location chosen by the Employer.

2. The load test shall be erected and performed by the VR Contractor at the VR Contractor's expense. The Employer shall pay for all costs associated with monitoring of the test by the FQCR.

3. The load test shall be performed in general accordance with ASTM D1143 Quick Test. The test foundation shall be loaded to 1.5 times the design load. Settlements of the test foundation shall be measured halfway between the center and each of the four corners. The average of the four readings shall be used to confirm acceptance of the required settlement criteria. The Contractor shall submit load test detail prior to load testing.

4. If uplift elements are used to develop the reaction load for the load test, they shall develop their uplift capacity below the depth being improved by the VRs or located horizontally at least 3.5 m away from the edge of the load test footing.

7.6 Restrictions

- VR construction is typically performed under the Site permit. The Contractor shall be responsible for obtaining any state and local permits (if required) and conforming to all state and local regulations.
- The Employer will be responsible for the precise delineation of all above and below ground utilities and obstructions
- The following shall also be listed within this section when applicable:
  - Environmental restrictions
  - Work boundaries
  - Hours for construction.

8. QUALITY CONTROL

The Contractor shall refer to the following.

MD-101-SMPM-PM-QA-PEI-0001 Contractor Quality Requirements